

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-3: Implement dynamic web applications by utilizing DOM-based client-side scripting and employing Java Servlets for server-side request processing and response handling. (L3)

Assessment Tool : Technical Presentation
Weightage: 20%

QUESTIONS:

Total Marks: 20

1. Servlet Architecture & Workflow

Question:

Present the **architecture of Java Servlets** and explain the complete **request–response cycle** with a neat diagram.

Focus Areas:

- Client–Server interaction
 - Role of Web Container
 - Request and Response objects
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2. Servlet Life Cycle Analysis

Question:

Explain the **Servlet Life Cycle** with a sequence diagram and analyze the role of `init()`, `service()`, and `destroy()` methods.

Focus Areas:

- Life cycle phases
 - Execution flow
 - Performance implications
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3. GET vs POST Method Evaluation

Question:

Compare **GET and POST methods** used in form submission. Present scenarios where each method is appropriate.

Focus Areas:

- Data transmission
 - Security aspects
 - URL limitations
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4. Session Tracking Techniques

Question:

Present different **session tracking mechanisms** (HttpSession, Cookies, URL rewriting).
Evaluate their advantages and limitations.

Focus Areas:

- Session persistence
 - Security concerns
 - Use-case comparison
-

5. Cookies vs Sessions – Critical Analysis

Question:

Analyze and compare **Cookies and Sessions** in web applications. Present real-time use cases.

Focus Areas:

- Storage location
 - Security risks
 - Performance considerations
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Code of Conduct:

I N. Meghana(192411129) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Technical Presentation					
Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

1. Servlet Architecture & Workflow

A Servlet is a Java program that runs on a web server and handles requests from users.

How it works

1. The client/browser sends a request to the server.
2. The request reaches the Web Container such as Tomcat.
3. The Web Container identifies the required servlet.
4. The servlet processes the request.
5. It uses the Request object to get information from the user.
6. It uses the Response object to send the result back.
7. The browser displays the response.

Simple flow:

Browser → Web Container → Servlet → Processing → Response → Browser

Main components

- Client: Sends the request.
- Web Container: Manages servlets and handles requests.
- Servlet: Processes the request.
- HttpServletRequest: Contains data sent by the client.
- HttpServletResponse: Used to send the response to the client.

Example:

When you submit a login form, the browser sends username and password to a servlet. The servlet checks the details and sends back a login success or failure message.

2. Servlet Life Cycle

The Servlet Life Cycle describes the different stages a servlet goes through from creation to removal.

There are three important methods:

1. init()

- Called only once when the servlet is created.
- Used for initialization.
- Example: establishing a database connection.

2. service()

- Called whenever a client sends a request.
- Processes the request and generates a response.

- It can handle different HTTP methods such as GET and POST.

3. destroy()

- Called once before the servlet is removed.
- Used to release resources.
- Example: closing a database connection.

Simple flow

Servlet Created → init() → service() → service() → service() → destroy()

The init() method runs once, while service() can run many times for different requests. This makes servlet processing efficient.

3. GET vs POST Method

GET and POST are commonly used for sending data from a client to a server.

GET	POST
Data is added to the URL	Data is sent in the request body
Data can be seen in the URL	Data is not normally shown in the URL
Suitable for searching	Suitable for login/form submission
Can be bookmarked	Usually not bookmarked with submitted data
Has URL length limitations	Can send larger amounts of data
Not suitable for sensitive data	Better choice for sensitive form data

GET example

A search request might look like:

example.com/search?name=phone

The search information appears in the URL.

POST example

For a login form, username and password are sent using POST rather than putting them in the URL.

Important: POST itself does not encrypt data. For real security, HTTPS should be used.

The assessment specifically asks you to compare data transmission, security aspects, and URL limitations.

4. Session Tracking Techniques

HTTP is generally stateless, meaning the server does not automatically remember a user's previous request.

Session tracking is used to maintain information about a user across multiple requests. The assessment focuses on three techniques: HttpSession, Cookies, and URL Rewriting.

1. HttpSession

The server creates a session for the user and stores information on the server.

Example:

A shopping website stores your cart information in your session while you browse different pages.

Advantage: Data is maintained on the server.

Limitation: Uses server memory/resources.

2. Cookies

A small piece of information is stored in the user's browser.

Example:

A website remembers your language preference.

Advantage: Reduces the need to store small pieces of information on the server.

Limitation: Users can disable or delete cookies.

3. URL Rewriting

The session ID is added to the URL to maintain the session.

Example:

website.com/products;jsessionid=ABC123

Advantage: Can work when cookies are disabled.

Limitation: The session ID appears in the URL, which can create security concerns.

5. Cookies vs Sessions

Both cookies and sessions are used to maintain user information, but they work differently.

Cookies

- Stored on the client/browser.
- Suitable for small amounts of information.
- Can remain after the browser is closed if configured as a persistent cookie.
- Can be deleted or disabled by the user.
- Example: Remembering a user's language or website preference.

Sessions

- Session data is maintained on the server.
- Usually associated with a session ID on the client.
- Useful for temporary user-specific data.

- Commonly used for login information and shopping carts.
- Ends when the session expires or is invalidated.

Simple comparison

Cookies	Sessions
Stored on client	Data stored on server
Good for small data	Good for user/session data
Can be disabled	Managed by server
Less server storage required	Uses server resources
Example: preferences	Example: shopping cart

Cookie = information kept in the browser.

Session = user information maintained by the server.